Information Processing and Management 63 (2026) 104339

Contents lists available at ScienceDirect

Information Processing and Management

journal homepage: www.elsevier.com/locate/ipm

X-SSL: Self-supervised X-ray threat detection with zero-shot and multi-modal learning

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ARTICLE INFO

Dataset link: https://github.com/lutao2021/PIDray, https://github.com/GreysonPhoenix/CLCXray

Keywords:
X-ray threat detection
Self-supervised learning (SSL)
Zero-shot object proposal
Multi-modal clustering
Knowledge distillation
Instance segmentation

ABSTRACT

Automated X-ray threat detection is challenged by cluttered baggage scans, severe object occlusions, and the scarcity of annotated datasets. Traditional supervised approaches are impractical, as they require large amounts of labeled data, which is difficult to obtain given the rarity of threat objects. Unsupervised methods, on the other hand, often fail to differentiate between threat and nonthreat items due to the complex grayscale nature and high object overlap inherent in X-ray imagery. To overcome these limitations, we propose **X-SSL** a novel self-supervised learning framework that eliminates manual annotations designed to perform threat localization. Our approach integrates spatial region extraction using MaskCut for zero-shot object proposal generation, contrastive multi-modal clustering that leverages both image and text encoders to cluster and label proposals into threat and nonthreat categories, and self-supervised knowledge distillation where a teacher–student model refines multiscale features from global and local image crops for improved representation learning. We evaluated X-SSL on two benchmark datasets: PIDray (39,000+ images) and CLCXray (14,000+ images), demonstrating significant improvements over previous state-of-the-art (SOTA) methods. In the hidden PIDray subset, X-SSL improves detection AP to 18.16 (+4.41 AP) and segmentation AP to 13.76 (+1.01 AP) over previous methods. On CLCXray, it achieves 39.26 detection AP (+5.93 AP) and 38.67 segmentation AP (+10.20 AP), significantly surpassing previous approaches. For classification, X-SSL achieves an accuracy of 43% on PIDray Hidden and 65% on CLCXray, further highlighting its superior performance compared to existing weakly supervised and unsupervised methods. Code will be available here: https://github.com/yonathan-kiflom/X-SSL.

1. Introduction

The growing complexities of modern security challenges have made automated threat detection systems indispensable in high-risk environments such as airports and cargo terminals. With millions of passengers and vast amounts of freight passing through checkpoints every day (National Research Council, 1996), X-ray imaging serves as a critical tool for uncovering concealed threats, including weapons, explosives, and other prohibited materials (Saavedra, Banerjee, & Mery, 2021). However, the increasing demand for faster and more accurate security screening is hindered by both technical and operational challenges, which limit the effectiveness of existing detection systems. X-ray scans often contain cluttered, overlapping, and occluded objects, making it challenging to

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https://doi.org/10.1016/j.ipm.2025.104339
Received 13 April 2025; Received in revised form 28 July 2025; Accepted 28 July 2025
Available online 9 August 2025
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distinguish threats from benign items (Uroukov & Speller, 2015). Threat objects may be hidden behind other objects or embedded within complex configurations. Moreover, X-ray images lack rich color and texture information, which hinders the performance of traditional object detection models such as Faster R-CNN (Ren, He, Girshick, & Sun, 2015), Mask R-CNN (He, Gkioxari, Dollár, & Girshick, 2017), and RetinaNet (Akçay, Kundegorski, Willcocks, & Breckon, 2018). These models, designed primarily for natural RGB images, struggle to generalize effectively to the unique characteristics of X-ray data (Hassan, Bettayeb, et al., 2020; Tao et al., 2021). An additional challenge lies in the limited availability of annotated X-ray datasets. Manually labeling X-ray images is time-consuming and impractical due to the rarity and diversity of threat items (Miao et al., 2019). The inherent class imbalance and data sparsity significantly hinder the training of robust deep learning models (Miao et al., 2019). Many security systems must therefore rely on heuristic algorithms or incomplete datasets, resulting in reduced reliability under real-world conditions. Furthermore, human operators conducting manual inspections often suffer from fatigue and cognitive overload, leading to error rates of up to 20% (Velayudhan, Hassan, Damiani, & Werghi, 2022). Automated threat detection has emerged as a promising alternative to address these limitations (Uroukov & Speller, 2015). Recent research has explored clutter-aware techniques (Wei et al., 2020), data augmentation for imbalanced classes (Ahmed et al., 2022), and tensor-based models for occlusion handling (Hassan, Akcay, Bennamoun, Khan, & Werghi, 2022; Hassan, Bettayeb, et al., 2020; Hassan, Shafay, et al., 2020; Velayudhan, Hassan, Ahmed, Damiani, & Werghi, 2022). While these methods have shown some improvements, they still require annotated data and often fail to adapt to new or unseen threat types. Therefore, there is a pressing need for scalable, annotation-free models that can handle the complexities of X-ray imagery.

To address these challenges, we propose self-supervised X-ray threat detection with zero-shot and multi-modal learning dubbed X-SSL. Our framework eliminates the need for annotated datasets by leveraging a three-stage process optimized for X-ray imagery. First, we implement a zero-shot object proposal extraction method using CutLER (Wang, Girdhar, Yu & Misra, 2023), which identifies regions of interest without any labeled data. Next, contrastive multi-modal clustering is employed to semantically differentiate threat and non-threat objects using both visual and textual cues. Finally, we utilize a self-supervised knowledge distillation strategy, where a teacher–student model architecture refines feature representations from global and local image crops, improving robustness and transferability for downstream tasks such as detection, segmentation, and classification. As shown in Fig. 1, x-SSL outperforms state-of-the-art (SOTA) methods across multiple benchmarks. For example, on the PIDray Hidden subset, our model demonstrates substantial improvements in both detection and segmentation performance over previous approaches. Similar performance gains are observed on the CLCXray dataset, confirming the model's adaptability to complex, real-world scenarios.

The central key characterizations of the proposed system are:

- We introduce a novel self-supervised framework designed specifically for X-ray threat detection, which completely removes the dependency on annotated datasets. This addresses the critical challenge of scarcity in labeled data and enables efficient and scalable training for real-world security applications.
- The framework employs a multi-modal approach to semantically cluster object proposals, effectively distinguishing threat and non-threat objects, while bypassing the need for manual annotations, thus significantly reducing the reliance on costly and time-consuming data labeling processes.
- A multi-stage approach is proposed, starting with zero-shot object proposal extraction using CutLER, followed by semantic clustering and feature refinement. This systematically addresses challenges like clutter, occlusion, and diversity in X-ray imagery, leading to robust feature extraction and accurate detection.
- Our proposed framework employs a distillation-based Self-Supervised Learning (SSL) strategy to refine and enhance feature representations, ensuring the extracted features are robust, transferable, and well-suited for downstream tasks such as detection and localization.

The remainder of this paper is organized as follows: Section 2 outlines the objectives of our project. Section 3 discusses related works on supervised and semi-supervised, and self-supervised and unsupervised threat localization in X-ray images. Transformer based feature extraction methods are also briefly discussed. Section 4 outlines the main components of our proposed framework with detailed explanation. Section 5 discusses implementation details, evaluation protocols employed and datasets used along with the proper explanation of the evaluation metrics used. Section 6 details the comparative results of all the downstream tasks employed and ablation study of each component of our proposed framework. Section 7 discusses the practical and theoretical implications of our proposed framework along with its limitations while noting the advancements over existing approaches. The conclusion of our project is provided in Section 8, along with the explanation of the abbreviations used in the paper and the necessary declaration statements taken during the course of the project.

2. Research objective

The primary objective of this research is to develop a SSL framework, X-SSL, for automated X-ray threat localization that eliminates the need for manual annotations while improving detection, segmentation and classification performances. To achieve this, we define the following specific objectives:

- Employ a novel teacher–student learning framework that refines multi-scale features from global and local image crops to enhance X-ray image understanding.
- Utilize MaskCut for unsupervised region proposal generation, enabling effective threat object localization without labeled data.

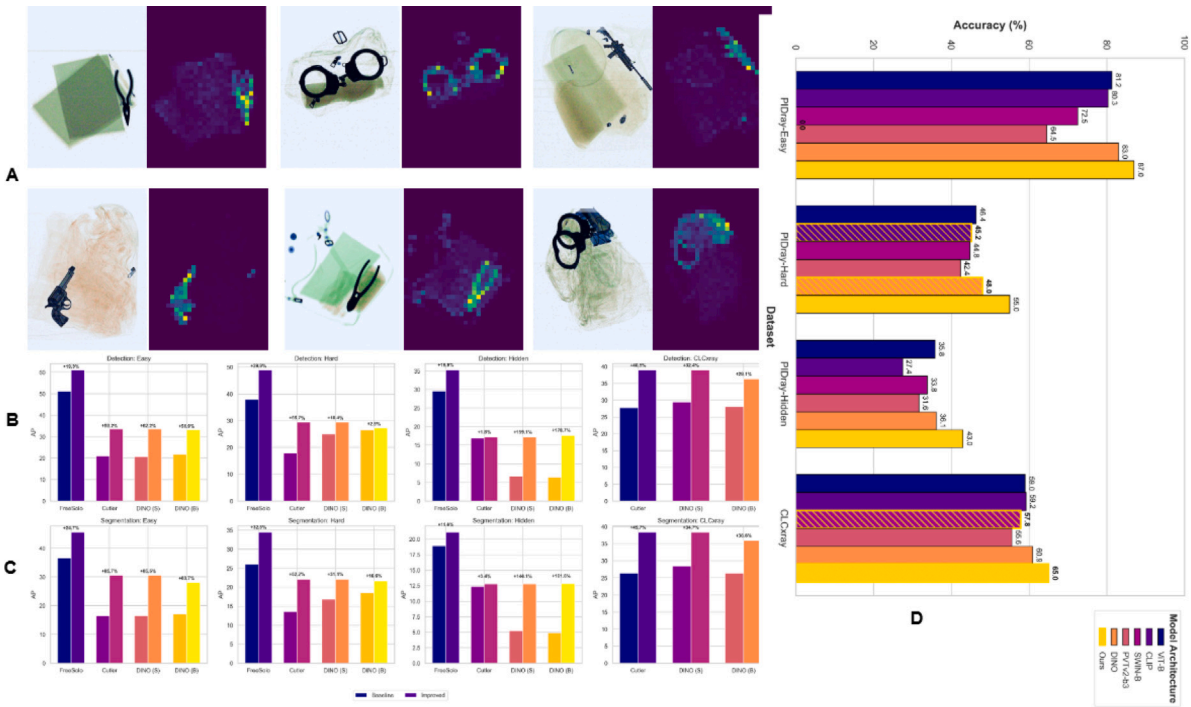

Fig. 1. **A.** Self-Attention from a Vision Transformer (ViT) trained with no supervision. The model provides the attention maps of the threat items detected in the X-ray baggage scans. The attention maps show X-SSL automatically learns the threat item in the baggage leading to a better object localization. **B.** Performance results on the detection downstream task across the datasets evaluated using AP metric. **C.** Performance results on the segmentation downstream task across the datasets evaluated using AP metric. **D.** Classification accuracy evaluated using several architectures across the datasets.

- Implement a contrastive multi-modal learning approach that leverages both image and text encoders to cluster and label proposals into threat and non-threat categories.
- Assess X-SSL on benchmark X-ray datasets (PIDray and CLCXray) by comparing detection, segmentation and classification performances against existing supervised and unsupervised methods.

3. Related work

Computer vision techniques have significantly advanced the field of X-ray threat localization, employing supervised (Shafay et al., 2022), semi-supervised (Akcay, Atapour-Abarghouei, & Breckon, 2018; Velayudhan et al., 2024), and unsupervised (Hassan, Akçay, Bennamoun, & Khan, 2023) learning paradigms tailored for this domain. While supervised methodologies rely on annotated datasets for object detection (Zhan, Shi, Liu, & Hu, 2023), self-supervised approaches leverage unlabeled data to mitigate the dependence on extensive manual annotation (Xiao, Zhou, Xie, Liu, & Huang, 2024). Convolutional Neural Networks (CNNs) (Wu, Qiu, Wang, & Feng, 2025) and Vision Transformers (ViTs) (Velayudhan, Hassan Ahmed, Hassan, et al., 2022) have demonstrated enhanced precision and generalizability in this context. Furthermore, specialized CNN architectures improve the interpretability of detection outcomes (Zhan et al., 2023), and hierarchical convolutional structures facilitate advanced feature extraction (Alansari et al., 2024; Hu, Chen, Huang, Qin, & Zheng, 2024).

3.1. Supervised and semi-supervised approaches

Supervised learning methods have been extensively employed for detecting concealed threats in X-ray imagery (Ahmed et al., 2023; Alansari et al., 2024). Recent methods such as MHT-X leverage transformer architectures to produce multi-level contour maps, significantly enhancing the detection of cluttered threats (Alansari et al., 2024). Similarly, Frequency and Edge-Aware Transformers (FEA-DETR) improve the detection accuracy by emphasizing occluded object boundaries using frequency-domain analysis (Hong, Zhou, & Xu, 2025). Additionally, programmable broad learning systems (Shafay et al., 2022) have been proposed for imbalanced datasets, enabling automatic optimization of network structures.

Semi-supervised frameworks further reduce the annotation burden. Approaches such as the background-adaptive Faster R-CNN employ adaptive feature extraction suitable for diverse operational scenarios (Sigman, Spell, Liang et al., 2020). Skip-GANomaly (Akçay, Atapour-Abarghouei, & Breckon, 2019) and GANomaly (Akcay et al., 2018) architectures effectively detect anomalies using minimal annotated data. Recently, a semi-supervised contour-driven broad learning system introduced by Velayudhan et al. (2024) significantly improved segmentation accuracy for concealed threats by explicitly incorporating boundary

information and dynamically focusing on critical regions. Moreover, attention mechanisms utilizing high-resolution semantic features further enhance segmentation tasks by effectively differentiating overlapping or occluded objects (Xia, Li, Huang et al., 2021). Additionally, few-shot learning methods tackle limited labeled data challenges for novel threats (Ahmed et al., 2024; Fang, Liu, Han et al., 2023).

Despite these advancements, supervised and semi-supervised methods face notable limitations. Supervised methods, such as those discussed by Alansari et al. (2024) and Hong et al. (2025), continue to heavily depend on annotated datasets, making large-scale deployment challenging due to high annotation costs and dataset scarcity (Shafay et al., 2022). Semi-supervised methods mitigate this to an extent but still require substantial weakly-labeled data, which can remain impractical in many operational contexts (Akcay et al., 2018; Akçay et al., 2019; Velayudhan et al., 2024). Moreover, current methods frequently struggle with generalizing to unseen threat types, especially in cluttered or occluded environments, often experiencing performance degradation under distribution shifts (Fang et al., 2023; Xia et al., 2021).

In contrast, our proposed framework, X-SSL, eliminates annotation requirements entirely by leveraging self-supervised learning (SSL). It integrates zero-shot object proposal extraction, multi-modal clustering, and knowledge distillation specifically tailored for X-ray images, addressing significant limitations of previous approaches and ensuring scalability and robust performance across diverse and unseen conditions.

3.2. Self-supervised and unsupervised approaches

SSL has emerged as a powerful alternative for reducing dependence on annotated datasets while achieving robust detection. Frameworks leveraging cross-domain and multi-modal data have demonstrated superior generalization capabilities, facilitating effective zero-shot identification and anomaly detection in previously unseen scenarios (Kerdvibulvech, 2025; Nasim, Akram, Khan et al., 2024). Hybrid frameworks integrating self-supervised and supervised methods have notably minimized manual annotation efforts while enhancing detection accuracy and generalization (Singh & Dhiraj, 2024). These frameworks leverage complementary strengths from labeled and unlabeled data to robustly detect anomalies and improve segmentation performance (Hassan et al., 2023). For instance, Griffin, Caldwell, Andrews, and Bohler (2018) demonstrated effective anomaly detection capabilities in cluttered baggage imagery using hybrid learning methods. Similarly, Andrews, Jaccard, Rogers, and Griffin (2017) successfully applied representation learning strategies to detect anomalies in complex X-ray cargo imagery, illustrating significant improvements in localization precision.

Despite these advancements, self-supervised and unsupervised methods encounter notable challenges. Many approaches depend on pretext tasks not optimized for the cluttered, grayscale, and occlusion-rich nature of X-ray images (Andrews et al., 2017; Griffin et al., 2018). Additionally, hybrid models often reintroduce annotation requirements, thereby reducing their scalability in practical deployments (Singh & Dhiraj, 2024). Inefficient feature extraction from high-dimensional X-ray data frequently results in sub-optimal detection performance, highlighting a critical gap in existing methodologies (Hassan et al., 2023).

Our proposed X-SSL framework directly addresses these limitations by entirely eliminating the need for manual annotations. Leveraging zero-shot object proposals, contrastive multi-modal clustering, and knowledge distillation specifically tailored for X-ray images, X-SSL enhances generalization to previously unseen threats and diverse operational conditions, providing a robust and scalable solution for real-world security screening scenarios.

3.3. Transformer-based feature extraction

ViTs have seen increasing adoption in SOTA models for unsupervised object detection (Cao, Joshi, Gui, & Wang, 2023; Caron et al., 2021; Siméoni et al., 2021; Wang, Girdhar, et al., 2023; Wang, Shen, et al., 2022). For example, DINO (self-DIstillation with NO labels) (Caron et al., 2021) utilizes a teacher–student framework wherein a ViT-based teacher network guides a student network to learn from unlabeled images, achieving notable performance in detection tasks. CutLER further extends this paradigm, employing ViTs for object segmentation by leveraging self-attention mechanisms to effectively capture long-range dependencies and contextual relationships, thereby significantly enhancing detection accuracy (Wang, Girdhar, et al., 2023).

Furthermore, HASSOD (Hierarchical Aggregation for Self-Supervised Object Detection) enhances robustness through the aggregation of multi-scale features, demonstrating particular effectiveness in detecting small and occluded objects (Cao et al., 2023). Similarly, LOST (LOcalizing Objects with Self-Supervised Transformers) refines attention maps to precisely localize objects without the requirement of manual annotations (Siméoni et al., 2021). TokenCut leverages the tokenization inherent in ViTs to segment images into discrete tokens, employing self-attention to efficiently isolate objects, particularly in high-resolution imagery (Wang, Shen, et al., 2022). These advancements collectively underscore the versatility and scalability of ViTs for feature extraction and unsupervised object detection (Caron et al., 2021; Siméoni et al., 2021; Wang, Girdhar, et al., 2023; Wang, Shen, et al., 2022).

Notwithstanding their efficacy in natural image domains, Transformer-based models such as DINO, CutLER, and LOST are primarily designed for natural RGB images and may not be optimally suited for the grayscale and cluttered characteristics prevalent in X-ray imagery (Siméoni et al., 2021; Wang, Girdhar, et al., 2023). Moreover, these methods often lack the semantic differentiation capabilities crucial for distinguishing threat objects in security-critical applications (Cao et al., 2023). In contrast, our proposed X-SSL framework builds upon CutLER's zero-shot object proposals and integrates multi-modal clustering and self-supervised distillation. This synergistic combination enables precise spatial and semantic localization of threats, specifically addressing the inherent complexities of X-ray security imaging.

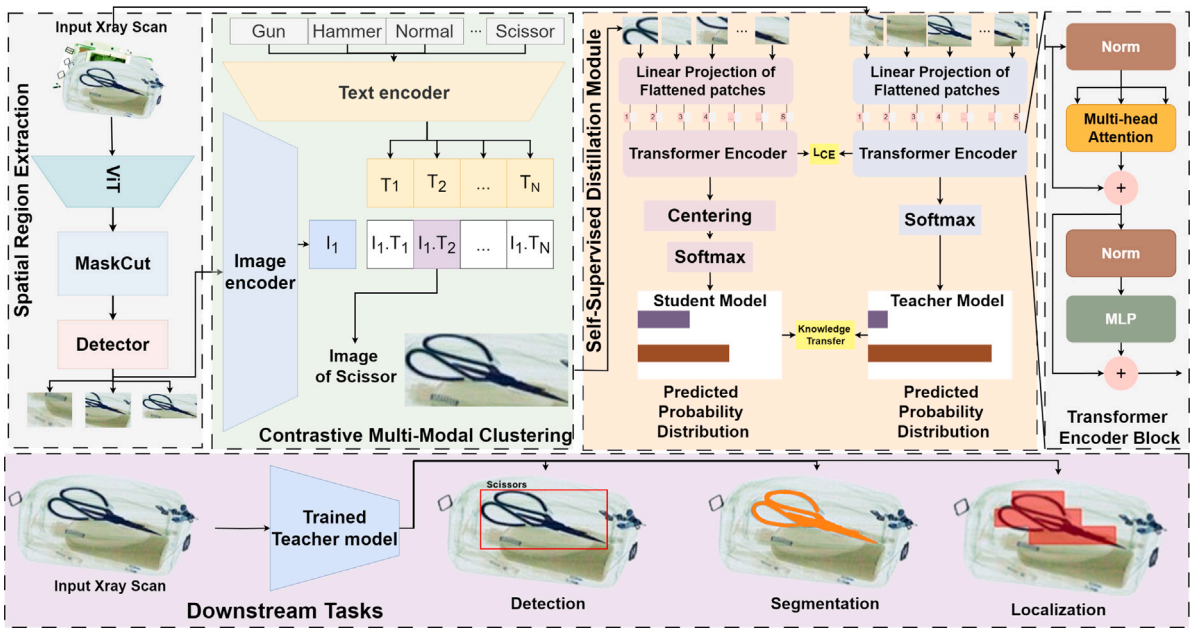

Fig. 2. Overview of the proposed X-SSL framework for self-supervised X-ray threat detection. The framework consists of three main stages: (1) Spatial Region Extraction, where zero-shot proposals are generated using MaskCut and a detector module, (2) Contrastive Multi-Modal Clustering, which employs image and text encoders to differentiate and semantically cluster object proposals into threat and non-threat categories, and (3) Self-Supervised Distillation, where a teacher–student architecture refines multi-scale feature representations for robust learning. The trained teacher model is applied to various downstream tasks, including detection, segmentation, and localization, demonstrating adaptability to cluttered X-ray imagery.

4. Proposed framework

In this work, we propose a self-supervised framework for threat and non-threat object detection in X-ray baggage imagery, specifically designed to address challenges related to label scarcity and complex object configurations, as illustrated in Fig. 2. The framework leverages both multi-modal and SSL techniques to extract robust, transferable features that enable accurate detection and classification. It comprises three primary stages. In the first stage, spatial region extraction is performed to generate object proposals from X-ray images. This stage uses a zero-shot object detection approach to ensure comprehensive coverage of all regions of interest, capturing both threat and non-threat objects. The proposals are further refined through segmentation techniques, which enhance the visibility and representation of objects in cluttered and occluded scenes. The second stage applies contrastive multi-modal clustering to semantically group the extracted proposals. Visual features from an image encoder are combined with textual embeddings from a text encoder to categorize each proposal, ensuring effective semantic differentiation between object types and improving classification performance. In the final stage, a self-supervised distillation module refines the feature representations of the clustered proposals. This module employs a teacher–student network architecture, where the student network aligns its predictions with the teacher's outputs by processing both global and localized views of the proposals. The distillation process improves feature robustness by reducing the influence of background noise and ensuring better generalization. The features extracted through this framework are highly adaptable and transferable, enabling efficient performance across various downstream tasks such as detection, segmentation, and localization.

4.1. Spatial region extraction

The proposal extraction module is a key component of our framework, serving as a critical step to identify and extract all potential regions of interest (ROIs) from the X-ray baggage images. This stage ensures comprehensive coverage by generating object proposals for all items present in the image, regardless of whether they are classified as threat or non-threat objects. To achieve this, we employ CutLER (Wang, Girdhar, et al., 2023), a zero-shot unsupervised object detector that leverages advanced detection algorithms to identify even subtle or occluded items within the X-ray scans. This is particularly advantageous in the context of X-ray imagery, where object visibility can be obscured by overlapping or cluttered items. The model effectively generates comprehensive object proposals, enabling downstream modules to process a diverse set of extracted regions. The extraction process is underpinned by MaskCut, a novel technique integrated into CutLER. MaskCut utilizes pre-trained self-supervised features to automatically generate multiple initial coarse binary masks $\mathbf{M}_i$ for each image I. Specifically, the process begins with the feature representation $\mathbf{F}$ of the image, derived as:

$$\mathbf{F} = \Phi(I), \tag{1}$$

where Φ represents a pre-trained self-supervised feature extractor. Next, MaskCut extends the Normalized Cuts (NCuts) algorithm for unsupervised segmentation, on a masked similarity matrix to extract multiple disjoint object regions. Given the affinity matrix $\mathbf{A}$ that captures the similarity between pixels or regions, the segmentation problem can be framed as:

$$\min_{\mathbf{y}} \frac{\mathbf{y}^T(\mathbf{D}-\mathbf{A})\mathbf{y}}{\mathbf{y}^T\mathbf{D}\mathbf{y}}, \tag{2}$$

where $\mathbf{D}$ is the degree matrix, and $\mathbf{y}$ denotes the segmentation labels. After each bipartition, the algorithm selects the foreground group using two constraint; it must include the patch with the highest eigenvector value and it should cover less than two of the image's four corners, reflecting a prior that the objects are centered. If these criteria are not met, the mask is flipped. For subsequent iterations, patches already marked as foreground are suppressed in the similarity matrix, allowing NCut to discover new objects. The proposal extraction ensures that multiple objects, including both normal and threat items, are effectively segmented and clustered, even when present in overlapping or cluttered arrangements. Each mask represents a potential object proposal, facilitating the detection of all items in the baggage, irrespective of their threat classification. This multi-mask approach generates a rich set of object proposals $\{P_1, P_2, \ldots, P_n\}$, where P_i represents a proposal corresponding to a specific region in the image. These proposals are subsequently passed to the next stage for semantic clustering and further refinement.

4.2. Contrastive multi-modal clustering

Following the extraction of object proposals, a pivotal challenge arises in effectively clustering these proposals in an unsupervised manner. Selecting a robust and efficient clustering approach is critical to the success of the framework, particularly given the diversity and complexity of X-ray imagery. To address this, a contrastive multi-modal methodology is adopted, leveraging both visual and textual modalities to enhance the clustering process. This dual-modality approach provides a richer semantic understanding, facilitating the effective differentiation of object categories. To augment the clustering process, categorical descriptions derived from the PIDRay (Wang, Zhang, Wen, Liu, & Wu, 2021) and CLCXray (Zhao, Zhu, Dou, Deng, & Wang, 2022) datasets were incorporated as textual prompts. These descriptions serve to contextualize the visual features extracted from the proposals, enabling a semantically grounded clustering mechanism. For each proposal P_i, the visual features $\mathbf{f}_i$ are obtained via a visual encoder Φ_v, and the textual embeddings $\mathbf{t}_c$ for class c are generated through a textual encoder Φ_t:

$$\mathbf{f}_i = \Phi_v(P_i), \quad \mathbf{t}_c = \Phi_t(T_c), \tag{3}$$

where T_c denotes the textual description associated with class c. To evaluate the alignment between the visual and textual embeddings, cosine similarity $S_{i,c}$ is employed as a metric:

$$S_{i,c} = \frac{\mathbf{f}_i \cdot \mathbf{t}_c}{\|\mathbf{f}_i\|\|\mathbf{t}_c\|}. \tag{4}$$

Each proposal P_i is subsequently assigned to the class c^* with the highest similarity score, formalized as:

$$c^* = \arg\max_c S_{i,c}. \tag{5}$$

This contrastive multi-modal clustering strategy exploits the interplay between visual and textual information, allowing the model to effectively disambiguate between diverse object categories. The textual descriptions provide semantic anchors that improve the clustering of proposals, particularly in scenarios involving visually ambiguous or cluttered X-ray imagery. By employing this approach, the framework achieves a robust and scalable clustering mechanism, addressing the limitations of uni-modal clustering methods and ensuring accurate categorization of all object proposals, including both normal and potential threat items.

4.3. Self-supervised distillation module

The Self-supervised Distillation Module is designed to extract robust visual features in a self-supervised manner, eliminating the need for labeled data. This method leverages two backbones, referred to as the student and teacher networks, which are designed to interact and iteratively refine each other's outputs in a knowledge distillation framework. Both networks receive distinct augmented views of the same image, facilitating the learning of meaningful representations through a self-supervised approach. The training process minimizes the discrepancy between the output distributions of the teacher and student networks. Given an input image I, two distinct augmentations are applied to generate global and local views. Global crops, denoted I_g, retain more than 80% of the original image content, while local crops, denoted I_l, encompass less than 80%. The student network processes a combination of both I_g and I_l, whereas the teacher network is provided only with I_g. The outputs of the teacher $T(I_g)$ and the student $S(I_g, I_l)$ are compared to guide the student network toward learning consistent feature representations. The objective function for this distillation process is defined as:

$$\mathcal{L}_{\text{distill}} = -\sum_c p_t^c \log p_s^c, \tag{6}$$

where p_t^c and p_s^c are the predicted probability distributions for class c from the teacher and student networks, respectively. The loss function ensures that the student's predictions align closely with the teacher's outputs.

Algorithm 1: X-SSL pseudocode

Require: Image dataset $\mathcal{D}$
Region Proposal Extraction:
For each image $I \in \mathcal{D}\}$ do:
  $\mathcal{R}_I \leftarrow \text{ExtractRegionProposals}(I)$
  Store $\mathcal{R}_I$
End For
Multi-Modal Clustering:
For each class name y_j:
  $e_j \leftarrow \text{TextEncoder}(y_j)$
End For
For each region proposal $r \in \bigcup \mathcal{R}_I$:
  $v_i \leftarrow \text{ImageEncoder}(r_i)$
  $c_i \leftarrow \arg\max_j \text{Similarity}(v_i, e_j)$
End For
Store clustered proposals F
Self-distillation:
#Student network `gs`, Teacher network `gt`, Center `C`
#Hyperparameters: Student/teacher temperatures `tps`, `tpt`; momentum rate `m`
#Initialization: `gt.params` $\leftarrow$ `gs.params`
Loader: $(\mathcal{D}, F)$
For each minibatch x in `loader`
  $x_1, x_2 \leftarrow \texttt{augment}(x), \texttt{augment}(x)$
  $s_1, s_2 \leftarrow \texttt{gs}(x_1), \texttt{gs}(x_2)$
  $t_1, t_2 \leftarrow \texttt{gt}(x_1), \texttt{gt}(x_2)$
  $\texttt{loss} \leftarrow \frac{1}{2}H(t_1, s_2) + \frac{1}{2}H(t_2, s_1)$
  `loss.backward()`
  `update(gs)`
  $\texttt{gt.params} \leftarrow l \cdot \texttt{gt.params} + (1 - l) \cdot \texttt{gs.params}$
  $\texttt{C} \leftarrow m \cdot \texttt{C} + (1 - m) \cdot \texttt{mean}(\texttt{cat}(t_1, t_2))$
End For
function $\text{H}(t, s)$
  $t \leftarrow \texttt{stop_gradient}(t)$
  $s \leftarrow \texttt{softmax}(s/\texttt{tps}, \texttt{dim} = 1)$
  $t \leftarrow \texttt{softmax}((t - C)/\texttt{tpt}, \texttt{dim} = 1)$
  return $-\texttt{mean}(\sum t \cdot \log s)$
end function

Our method incorporates pre-processed object proposals instead of random crops for the local views. These proposals are generated from the preceding spatial region extraction module, ensuring that the local views contain regions more relevant to the target object. By feeding the student network with pre-processed proposal images, the distribution graphs generated by the student and teacher models more accurately represent the underlying data. This refinement offers two key advantages. First, it removes unnecessary background noise present in random local crops, focusing the learning process on the relevant features. Second, the alignment between global and local views improves, resulting in enhanced target localization. This targeted approach ultimately leads to superior performance compared to methods that rely on random local sections. The pseudocode for the complete framework is provided in Algorithm 1.

4.4. Downstream tasks

The proposed model serves as a critical foundation for various downstream tasks in X-ray threat detection and localization. By leveraging the robust feature representations learned through self-supervised training, the model provides pretrained weights and adaptable features that significantly enhance the performance of tasks such as object detection, instance segmentation, and classification. In object detection, the model facilitates the efficient identification and localization of potential threats, including concealed or prohibited items within X-ray scans. The use of robust feature embeddings enables accurate detection even under challenging conditions, such as when threat objects are partially obscured, occluded, or appear in diverse orientations. For instance segmentation, the model supports the precise delineation of threat objects by providing pixel-level boundaries. This is particularly beneficial in high-density baggage screening environments, where objects are often tightly packed or overlapping. Accurate segmentation enhances the system's ability to separate and analyze multiple items within a single scan, reducing false

positives and improving interpretability. In classification tasks, the model enables reliable categorization of detected objects by distinguishing between benign and potentially hazardous items.

5. Experimental setup

In this section, we present the experimental details utilized to assess our proposed model. We initiate with a summary of the datasets employed, specifically PIDray and CLCxray. We then delve into technical implementation details specifics of our approach, covering aspects such as the computational resources, training protocols, selected SOTA methods, and strategies for data pseudo-labeling. Lastly, we discuss the performance metrics used to assess the proposed X-SSL.

5.1. Datasets

The proposed self-supervised X-SSL model was evaluated on two large-scale publicly available X-ray datasets: PIDray and CLCXray. These datasets, designed for security screening applications, provide diverse and realistic imagery to benchmark detection, segmentation, and classification performance.

CLCXray: The CLCXray (Zhao et al., 2022) dataset consists of 9,565 X-ray images, divided into two subsets. The first subset includes 4,543 real images captured in operational subway security checkpoints, while the second subset comprises 5,022 synthetic images created by scanning manually designed baggage setups. The dataset includes 12 distinct categories of prohibited items. Among these, five categories represent cutting instruments, such as blades, knives, scissors, and Swiss Army knives. The remaining seven categories cover various types of liquid containers, including cans, glass bottles, plastic bottles, vacuum cups, spray cans, and tins. Annotations are provided in COCO format, which contains detailed object information such as bounding boxes, category labels, and segmentation masks, enabling standardized evaluation across object detection and segmentation tasks (Zhao et al., 2022).

PIDray: PIDray (Wang, Zhang, et al., 2021) is a large dataset specifically created for detecting prohibited items in security applications. It contains a total of 124,486 X-ray images categorized into 12 distinct object types, including firearms, knives, scissors, handcuffs, hammers, lighters, power banks, and ammunition. These images are annotated following the COCO format at both the image and instance levels. The dataset is divided into training and testing subsets, with 29,457 images reserved for training and 18,220 images used for testing. The test set is further organized into three levels of detection complexity: the “easy” subset features images with a single visible threat item and minimal background clutter; the “hard” subset increases complexity by including multiple threat items arranged in crowded, overlapping scenes; and the “hidden” subset presents the most challenging scenarios, where threat objects are intentionally concealed behind other items to simulate real-world occlusion, dense clutter, and visual ambiguity. These progressively difficult subsets form the basis for evaluating model robustness under escalating levels of detection difficulty.

5.2. Implementation details

All experiments were conducted on a high-performance workstation equipped with an Nvidia GeForce RTX 3080 GPU (8 GB VRAM), 32 GB of RAM, and an Intel processor. We evaluated our model on three downstream tasks: object detection, instance segmentation, and single-label classification. The model’s training pipeline utilized standard deep learning frameworks with appropriate data augmentation and optimization techniques to ensure robust performance.

5.3. Evaluation metrics

To evaluate the effectiveness of the proposed framework, we adopted standard evaluation metrics commonly used in object detection, instance segmentation, and classification tasks. For object detection and segmentation, we report Average Precision (AP), which summarizes the precision–recall curve across multiple IoU thresholds, and AP50, which measures precision at a fixed Intersection over Union (IoU) threshold of 50%. These metrics reflect the model’s ability to accurately localize and segment objects. For classification tasks, we use Accuracy, Precision, Recall, and F1-score to provide a comprehensive assessment of prediction quality, balancing both correctness and completeness. These metrics enable direct and meaningful comparison against existing SOTA methods across all tasks.

5.4. Downstream evaluation protocol

To assess the generalization capability of our proposed framework, we conducted a comprehensive downstream task evaluation on single-label classification, object detection, and instance segmentation. To form a structured evaluation protocol, we followed three approaches for comparison and each approach followed a fine-tuning process of the model on the train set of the dataset, and evaluated on the test set in question. Our first approach utilized the initial provided weights of the chosen downstream task model. This is essential in having the baseline of the model’s performance on downstream tasks of X-ray threat images. For our framework’s evaluation against previous SOTA framework employed in our experimental analysis, DINO (Caron et al., 2021), we employed the provided weights of both frameworks on the chosen downstream task model for benchmarking. This is essential in having a unified and consistent comparative results of both frameworks under the same conditions.

5.5. SOTA methods for comparison

We evaluated the performance of our proposed X-SSL framework across three key downstream tasks relevant to X-ray threat detection: single-label classification, object detection, and instance segmentation. All evaluations followed a consistent fine-tuning protocol on the training split of each dataset and were tested on the corresponding test set, as detailed in Section 5.4. For each task, we selected SOTA models representative of current best practices in supervised, self-supervised, and unsupervised vision learning.

Single-label Classification: To benchmark classification performance, we compared X-SSL with two foundation models and three ViT architectures:

- CLIP (Radford et al., 2021): A vision–language model that learns to connect images and text by training on a large dataset of image–text pairs. It works by embedding both images and text into a shared space where related pairs are close together, enabling tasks like zero-shot image classification without needing task-specific training.
- DINO-B (Caron et al., 2021): A SSL method for training ViTs or CNNs without labeled data. It works by having two neural networks (a student and a teacher) learn to produce similar representations of different augmented views of the same image, enabling the model to learn meaningful features useful for tasks like classification, detection, or segmentation.
- ViT-B (Dosovitskiy et al., 2021): A neural network architecture that applies the Transformer model, originally designed for language, to image data. It splits an image into patches, embeds them like tokens, and processes them using self-attention, enabling it to capture global relationships in the image without using convolutions.
- Swin-B (Liu et al., 2021): A hierarchical ViT that processes images using non-overlapping local windows, with a shifting mechanism to enable cross-window connections. This design makes it efficient and scalable for high-resolution vision tasks like object detection and segmentation, while still capturing both local and global features.
- PVTv2-B3 (Wang et al., 2022): An improved version of the original PVT (Wang, Xie, et al., 2021), designed for dense prediction tasks like object detection and segmentation. It uses a pyramid structure to handle multi-scale features and incorporates optimizations like linear attention to reduce computational cost while maintaining high accuracy on vision tasks.

Object Detection and Instance Segmentation: For evaluating object detection performance in cluttered and occluded X-ray imagery, we selected strong unsupervised and self-supervised baselines that represent the current SOTA in zero-shot and label-free object localization:

- LOST (Siméoni et al., 2021): A fully unsupervised object localization method that uses transformer-based models (like DINO) to identify objects in images without any labels. It analyzes self-attention maps to find coherent regions (i.e., likely objects) based on how strongly image patches relate to each other, allowing it to localize objects without supervision or bounding box annotations.
- CutLER (Wang, Girdhar, et al., 2023): A recent unsupervised object detection and instance segmentation method that uses self-supervised features from vision transformers (like DINO) without needing labeled data. It first generates rough object masks using a technique called MaskCut, then trains a detector through self-training with a robust loss to refine predictions. CutLER achieves strong performance in zero-shot and few-shot settings across diverse visual domains.
- CutLER + Fine-tuned DINO (Caron et al., 2021; Wang, Girdhar, et al., 2023): This variant combines strong self-supervised features from a DINO model (pretrained and optionally fine-tuned on a domain) with CutLER's unsupervised object detection pipeline. The fine-tuned DINO provides more domain-relevant feature representations, which improves the quality of object masks generated by MaskCut. This enhanced input boosts the overall performance of CutLER, especially in domain-specific or few-shot settings.

6. Experiments and results

Our proposed X-SSL framework is evaluated on the aforementioned datasets and compared against recent SOTA models on three major downstream tasks. Comparative results and ablation results are explained below in detail.

6.1. Comparative results

In this section, we evaluated our framework on the three major downstream tasks against recent SOTA models and provided analysis of our findings.

6.1.1. Classification and localization

Following the protocol on Section 5.4 and comparing with SOTA methods in Section 5.5, we evaluate our model's performance on single-label classification downstream task on both the PIDray (Wang, Zhang, et al., 2021) and CLCXray (Zhao et al., 2022) datasets. As shown in Table 1, our model achieved the highest performance across most test sets, surpassing existing methods by notable margins in Accuracy, Precision, Recall, and F1-Score. However, to keep the performance comparison fair and consistent, we will focus on accuracy results as it is the most commonly reported metric on SOTA models. On the PIDray dataset (easy test set), our model achieved an accuracy of 87%, marking a +4% improvement over the best-performing baseline. In harder test scenarios, including the hidden set, our model maintained a significant advantage, achieving a +6.86% accuracy improvement. Similarly, on the CLCXray dataset, our model outperformed others, achieving 65% accuracy and demonstrating the robustness of our classification

Table 1
Single label classification results comparing various models on PIDray and CLCXray datasets. Metrics include Accuracy, Precision, Recall, and F1-Score across different difficulty levels. Improvement indicates the difference between our method and the previous best-performing approach, computed as a simple subtraction of metric values.

Model	PIDray												CLCXray			
	Easy				Hard				Hidden				Overall			
	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
ViT-B Dosovitskiy et al. (2021)	81.24	83	81	80	46.36	48	46	46	35.80	48	36	32	58.96	55	59	54
CLIP Radford et al. (2021)	80.26	81	80	79	45.20	47	45	44	27.45	40	27	21	59.21	60	59	55
SWIN-B Liu et al. (2021)	72.52	72	73	70	44.75	47	45	43	33.83	**54**	34	29	57.83	50	58	52
PVTv2-b3 Wang et al. (2022)	64.54	66	65	63	42.37	43	42	41	31.65	45	32	28	55.56	47	56	50
DINO Caron et al. (2021)	83	84	82	82	48	50	48	47	36.14	48	38	35	60.87	57	62	55
Ours	**87**	**86**	**88**	**87**	**55**	**53**	**54**	**53**	**43**	52	**53**	**51**	**65**	**60**	**65**	**61**
Improvement	+4.0	+2.0	+6.0	+5.0	+7.0	+3.0	+6.0	+6.0	+6.86	−2.0	+15.0	+16.0	+4.13	+0.0	+3.0	+6.0

approach even in challenging real-world conditions with high occlusion and clutter. In addition to quantitative evaluation, we assess localization quality by visualizing feature maps from the self-attention layers of ViT-based models. Fig. 3 presents a qualitative comparison between our method and DINO (Caron et al., 2021), the second-best performing approach and the most comparable self-supervised baseline. Both models generate attention-based localization maps, allowing us to visually examine their capacity to focus on threat-relevant regions. The figure highlights X-SSL's stronger ability to concentrate on critical areas, supporting its improved classification performance.

It is important to contextualize the 43% accuracy on the PIDray Hidden subset. While this number may appear modest in isolation, it must be interpreted in light of the extreme difficulty of the Hidden split. As discussed earlier, these images are designed to simulate high-stakes real-world scenarios where threat objects are heavily occluded, densely packed among benign items, and often indistinguishable due to limited semantic cues in pseudo-colored X-ray imagery. Additionally, the classification task is formulated as single-label, despite the presence of multiple overlapping objects in each image, which adds substantial ambiguity to the learning objective. Moreover, our model is trained entirely through self-supervised learning, without access to any labeled data during pretraining, placing it at a disadvantage compared to models initialized with semantic-rich supervision (e.g., ImageNet-pretrained baselines). Despite these compounded challenges, our method not only surpasses strong self-supervised baselines like DINO (Caron et al., 2021) by +6.86%, but also outperforms vision–language foundation models like CLIP (Radford et al., 2021), which benefit from large-scale paired text-image data. Achieving SOTA results under such adverse conditions confirms the robustness and transferability of our learned representations and highlights the practical value of our approach in high-risk detection tasks where data is scarce and occlusion is unavoidable.

6.1.2. Object detection

Following the evaluation protocol outlined in Section 5.4 and the SOTA baselines introduced in Section 5.5, we assess our model's performance on the object detection task using the PIDray (Wang, Zhang, et al., 2021) and CLCXray (Zhao et al., 2022) datasets. The results shown in Table 2 demonstrate that our model consistently outperformed competing approaches across various datasets and test set complexities (easy, hard, hidden). On the PIDray dataset, our model achieved an AP of 33.41% with the ViTSmall architecture on the easy test set, surpassing the nearest competitor by +7.75%. For the hard and hidden test sets, our model improved AP by +4.58% and +0.30%, respectively. In comparison to models utilizing pretrained weights, our approach demonstrated superior generalization and object localization capabilities, particularly in cluttered and occluded regions. On the CLCXray dataset, our method achieved an AP of 38.97%, with improvements of +5.65% over previous results. These gains highlight the effectiveness of our multi-modal clustering and feature extraction techniques in real-world threat detection scenarios.

6.1.3. Instance segmentation

Instance segmentation evaluations were conducted using the same protocols as object detection. Our model was benchmarked against SOTA models detailed in Section 5.5. Table 2 presents the results, which confirm the robustness of our proposed method in segmenting threat and non-threat objects from X-ray scans. Our model achieved an AP of 30.55% on the PIDray easy test set with ViTSmall, representing an improvement of +14.08% over previous best results. Similarly, on the hard and hidden sets, we observed improvements of +5.22% and +0.42%, respectively. For the CLCXray dataset, our approach attained an AP of 38.36%, outperforming existing methods by +9.63%. The consistent performance across all test sets underscores our model's ability to accurately segment complex scenes with overlapping or obscured objects, making it a valuable solution for real-world security screening applications.

6.1.4. Discussion

As discussed above, our model outperforms the SOTA models across the benchmarked datasets on most metrics employed and is competitive on the others. A key factor in contributing to the superior performance is the integration of region proposals as local crops within the self-supervised training pipeline. Unlike previously implemented approaches that rely on randomly sampled local

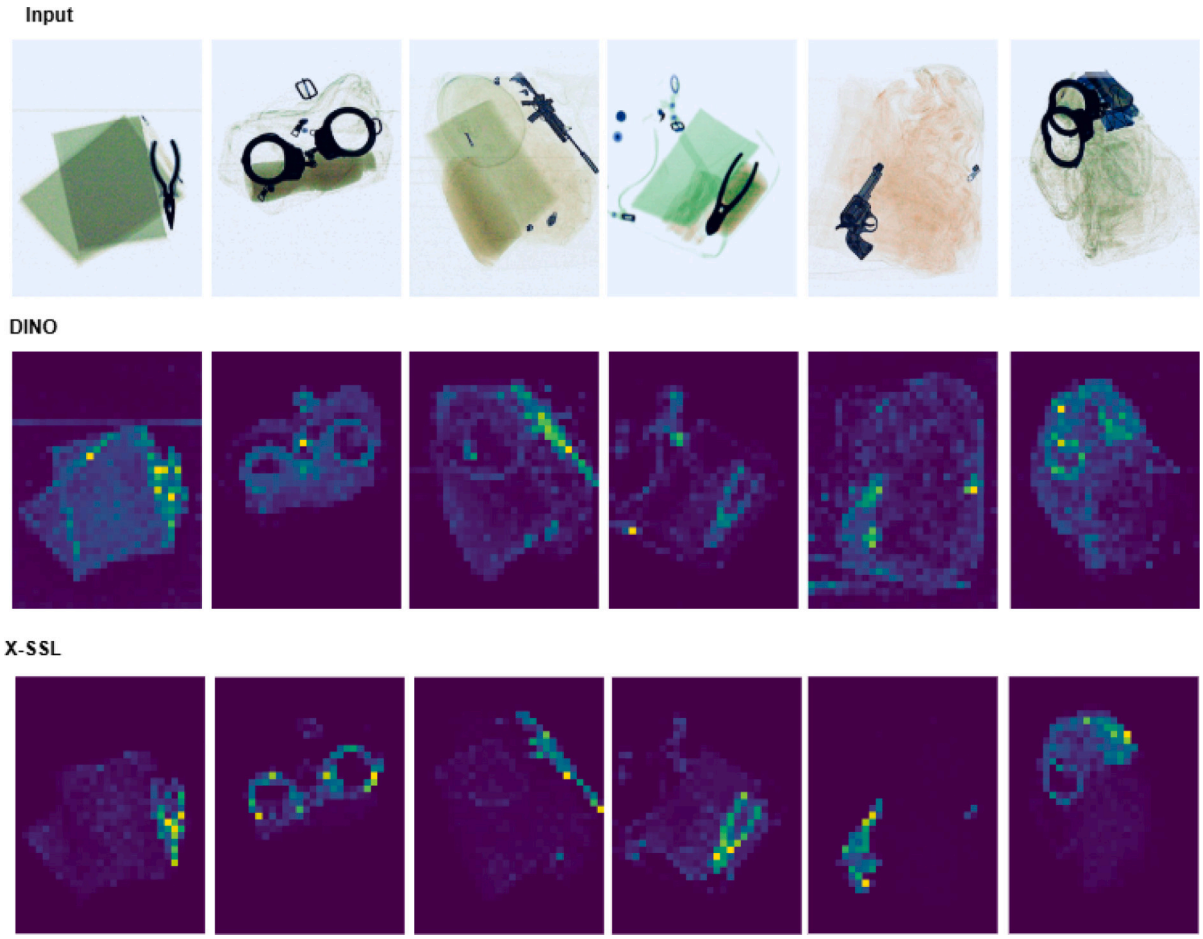

Fig. 3. Attention maps extracted using DINO model and our proposed X-SSL model. Both models are trained on X-ray threat baggage images with no supervision and outputs illustrated for visualization.

Table 2
Comparison table on downstream tasks. Metrics include AP and AP50 across different difficulty levels. Det represents Object Detection, and Seg represents Instance Segmentation. Improvement indicates the performance gain of our method over the previous best-performing approach, computed via simple metric-wise subtraction.

Model	Architecture	Easy				Hard				Hidden				CLCxray			
		Det		Seg		Det		Seg		Det		Seg		Det		Seg	
		AP	AP50	AP	AP50	AP	AP50	AP	AP50	AP	AP50	AP	AP50	AP	AP50	AP	AP50
FreeSOLO Wang, Yu, et al. (2022)	ResNet	51.17	67.39	36.52	65.79	37.94	63.78	25.97	56.55	29.62	43.34	18.90	40.36	–	–	–	–
FreeSOLO +Ours	ResNet	**61.03**	**76.77**	**45.53**	**75.60**	**48.89**	**73.13**	**34.42**	**67.95**	**35.23**	**51.55**	**21.09**	**46.26**	–	–	–	–
Improvement	—	+9.86	+9.38	+9.01	+9.81	+10.95	+9.35	+8.45	+11.40	+5.61	+8.21	+2.19	+5.90	–	–	–	–
LOST Siméoni et al. (2021)	ViTSmall	25.66	**61.79**	–	–	14.88	47.34	–	–	7.38	21.01	–	–	33.32	**64.23**	–	–
CutLER (pretrained) Wang, Girdhar, et al. (2023)	ViTSmall	20.98	31.79	16.45	30.11	17.80	29.22	13.58	24.55	16.88	24.47	12.35	21.97	27.74	40.08	26.33	39.73
CutLER +finetuned DINO Wang, Girdhar, et al. (2023) Caron et al. (2021)	ViTSmall	20.60	32.95	16.47	30.13	24.91	40.15	16.80	32.83	6.63	13.74	5.19	12.50	29.44	46.68	28.47	46.65
CutLER +Ours	ViTSmall	**33.41**	51.15	**30.55**	**49.73**	**29.49**	**49.27**	**22.02**	**42.60**	**17.18**	**27.79**	**12.77**	**25.25**	**38.97**	55.13	**38.36**	**55.17**
Improvement	—	+7.75	−10.64	+14.08	+19.60	+4.58	+1.93	+5.22	+9.77	+0.30	+3.32	+0.42	+3.28	+5.65	−9.10	+9.89	+8.52
CutLER +finetuned DINO Wang, Girdhar, et al. (2023) Caron et al. (2021)	ViTBase	21.88	34.77	17.15	31.31	26.56	42.33	18.53	34.98	6.33	13.11	4.90	12.10	28.15	45.38	26.33	45.45
CutLER +Ours	ViTBase	**33.24**	**52.18**	**28.08**	**48.75**	**27.31**	**48.14**	**21.61**	**41.20**	**17.64**	**29.80**	**12.82**	**26.86**	**36.33**	**53.16**	**35.96**	**53.10**
Improvement	—	+11.36	+17.41	+10.93	+17.44	+0.75	+5.81	+3.08	+6.22	+11.31	+16.69	+7.92	+14.76	+8.18	+7.78	+9.63	+7.65

views, our method leverages a more precise and semantically clustered meaningful proposals derived in the first two stages of our framework. By feeding these localized region proposals into the student network, we are guiding it to form a more target-oriented approach by reducing the background noise, particularly in cluttered X-ray scans. It ensures that the student network is explicitly trained to focus on actual object regions, resulting in improved localization capabilities. Consequently, the model achieves a clearer learning signal and reduced background noise, all of which results superior or competitive performance on downstream tasks.

Table 3
Comparison of performance with and without the multiple descriptions protocol on PIDray. The proposed protocol demonstrates significant improvements over the alternative protocol across all tasks and evaluation subsets. Improvement indicates the performance gain of our method over the previous best-performing approach, computed via simple metric-wise subtraction.

Model	Architecture	Easy				Hard				Hidden			
		Det		Seg		Det		Seg		Det		Seg	
		AP	AP50	AP	AP50	AP	AP50	AP	AP50	AP	AP50	AP	AP50
Multiple Caption	ViTSmall	22.06	34.96	17.79	31.95	18.52	31.21	13.09	25.16	12.31	20.14	9.08	18.34
Ours	ViTSmall	**33.41**	**51.15**	**30.55**	**49.73**	**29.49**	**49.27**	**22.02**	**42.60**	**17.18**	**27.79**	**12.77**	**25.25**
Improvement	—	+11.35	+16.19	+12.76	+17.78	+10.97	+18.06	+8.93	+17.44	+4.87	+7.65	+3.69	+6.91
Multiple Caption	ViTBase	20.98	33.61	16.72	30.54	17.67	28.79	14.44	26.29	14.93	24.57	11.63	22.64
Ours	ViTBase	**33.24**	**52.18**	**28.08**	**48.75**	**27.31**	**48.14**	**21.61**	**41.20**	**17.64**	**29.80**	**12.82**	**26.86**
Improvement	—	+12.26	+18.57	+11.36	+18.21	+9.64	+19.35	+7.17	+14.91	+2.71	+5.23	+1.19	+4.22

Table 4
Comparison of multi-modal and KNN clustering techniques on the PIDray and CLCXray datasets. The table shows accuracy, precision, and recall values for each dataset. Improvement indicates the performance gain of our method over the KNN, computed via simple metric-wise subtraction.

Model	PIDray			CLCxray		
	Acc	Pre	Rec	Acc	Pre	Rec
KNN	56.7	47.7	72.1	60.5	45.1	73.6
Ours	**72.4**	**64.8**	**83.7**	**68.3**	**75.6**	**84.5**
Improvement	+15.7	+17.1	+11.6	+7.8	+30.5	+10.9

6.2. Ablation study

To better understand the contribution of each component in our framework, we conduct a series of ablation experiments by systematically modifying individual modules, we assess their performance on X-ray scanned dataset and validate the design choices of our approach.

6.2.1. Impact of multiple descriptions in clustering

To better understand the impact of various design choices within the proposed X-SSL framework, we conducted an ablation study to evaluate the effect of providing multiple textual descriptions for each class during the clustering phase. This alternative protocol aimed to enhance the semantic space by introducing greater contextual diversity for object proposals. For each proposal, the model computed probabilities against all available descriptions for a given class and assigned the class with the highest probability. The goal was to leverage richer semantic information to improve clustering accuracy and overall performance. However, contrary to expectations, this approach led to a decrease in performance across detection and segmentation tasks. The inclusion of multiple descriptions increased semantic ambiguity, likely causing the model to struggle with over-representation and misclassification of objects. As shown in Table 3, the performance degradation is evident across all test sets (easy, hard, and hidden) when compared to the baseline protocol. Specifically, on the PIDray Hidden subset, the multiple description protocol using ViTSmall architecture achieved a detection AP of 12.31 and segmentation AP of 9.08, whereas the original protocol significantly outperformed it with detection and segmentation AP scores of 17.18 and 12.77, respectively. This pattern was consistent across both ViTSmall and ViTBase architectures, with improvements of up to 12.26 AP in detection and 18.57 AP50 in segmentation on the easy test set when using our baseline approach.

6.2.2. Clustering effectiveness analysis

An important aspect of the proposed X-SSL framework is its multi-modal clustering mechanism, which leverages both visual and textual embeddings to enhance the semantic grouping of object proposals. To assess the effectiveness of this approach, we conducted a comparative study against a conventional K-Nearest Neighbors (KNN)-based clustering technique. Table 4 presents a detailed comparison of the clustering methods in terms of accuracy, precision, and recall across the PIDray and CLCXray datasets. The results demonstrate that multi-modal clustering consistently outperforms KNN clustering across all metrics, with improvements of over 15% in accuracy and significant gains in precision and recall. For example, on the PIDray dataset, the multi-modal clustering approach achieved an accuracy of 72.4% compared to 56.7% for KNN. Fig. 4 provides visual examples that highlight the qualitative differences between the two clustering techniques. The multi-modal approach shows a greater capacity to distinguish between visually similar but semantically distinct objects, such as handcuffs and scissors. In contrast, KNN clustering exhibits higher confusion, often misclassifying objects with overlapping shapes and textures due to its reliance on local feature similarity alone. This suggests that incorporating textual context into the clustering process enhances the model's ability to handle occluded and cluttered scenes, improving both classification and segmentation performance.

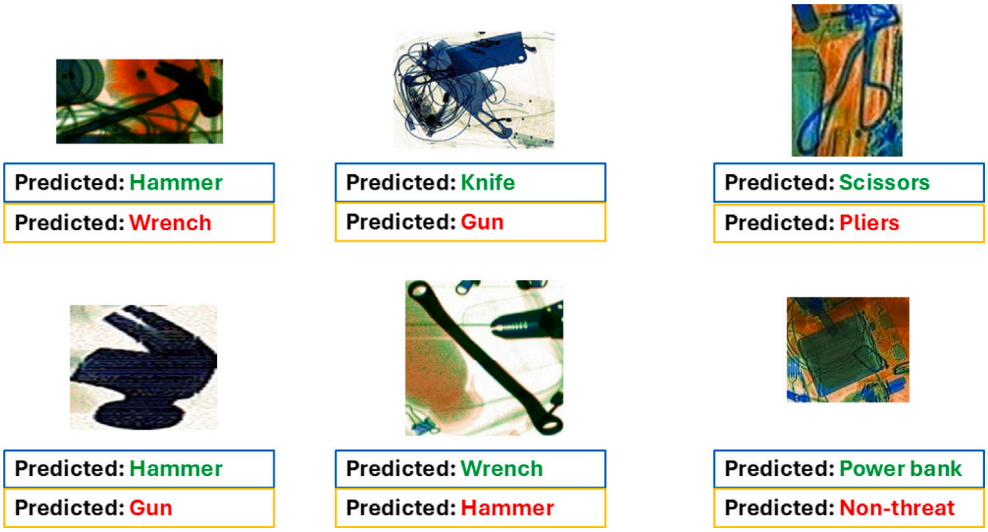

Fig. 4. Comparison of multi-modal clustering and KNN clustering for selected objects. The multi-modal approach predictions (enclosed by blue box demonstrate better semantic consistency, particularly in distinguishing similar objects under occlusion and clutter. KNN clustering prediction are enclosed by yellow box.

Table 5
Quantitative comparison of feature extraction modules across different clustering techniques on the PIDray subsets and CLCXray datasets. The table shows mAP values for each dataset. Improvement indicates the performance gain of our method over the previous best-performing approach, computed via simple metric-wise subtraction.

Model	PIDray			CLCxray
	Easy	Hard	Hidden	
LOST Siméoni et al. (2021)	17.6	12.1	9.4	13.7
FreeSOLO Wang, Yu, et al. (2022)	19.8	14.2	12.3	16.4
Ours	**22.1**	**17.9**	**13.5**	**23.6**
Improvement	+2.3	+5.6	+1.2	+7.2

6.2.3. Region extraction effectiveness

To evaluate the effectiveness of unsupervised region proposal extraction in X-ray images, we compared three recent advanced approaches. MaskCut (Wang, Girdhar, et al., 2023) leverages normalized cuts on self-supervised ViT features, enabling multi-object extraction through iterative bipartitioning of a masked affinity matrix. This allowed us to extract multiple precise object boundaries even under occlusion or clutter, which is critical in our case. In contrast, LOST (Siméoni et al., 2021) relies on saliency maps to produce a single object proposal and often failing in scenes with overlapping items. FreeSOLO (Wang, Yu, et al., 2022) performs better by predicting object masks via contour tracing, but its reliance on dense visual cues causes it to merge overlapping or adjacent items in occluded scenes.

Fig. 5 illustrates visual examples demonstrating the performance of these methods in extracting regions of interest from X-ray baggage scans containing both threat and non-threat objects. Complementing this, Table 5 presents a quantitative comparison across the PIDray and CLCXray datasets. Our approach outperformed both baselines on all test subsets. Specifically, we achieved the highest mean Average Precision (mAP) scores, with notable improvements of +2.3% on the PIDray easy set, +5.6% on the hard set, +1.2% on the hidden set, and a substantial +7.2% on the CLCXray dataset. These results affirm the superiority of our region proposal pipeline in extracting accurate and robust object candidates, which is critical for downstream classification and detection tasks.

7. Discussion

The findings of this study demonstrate that SSL and multi-modal contrastive learning can significantly improve X-ray threat detection in cluttered and occluded baggage scans. The proposed X-SSL framework builds upon and surpasses prior weakly supervised and unsupervised methods, addressing fundamental challenges in security screening research.

7.1. Practical implications

Our project has significant practical implications for security screening applications. By reducing the dependency on manual annotations, our approach enhances scalability and cost-effectiveness, making it more viable for real-world deployment. In addition,

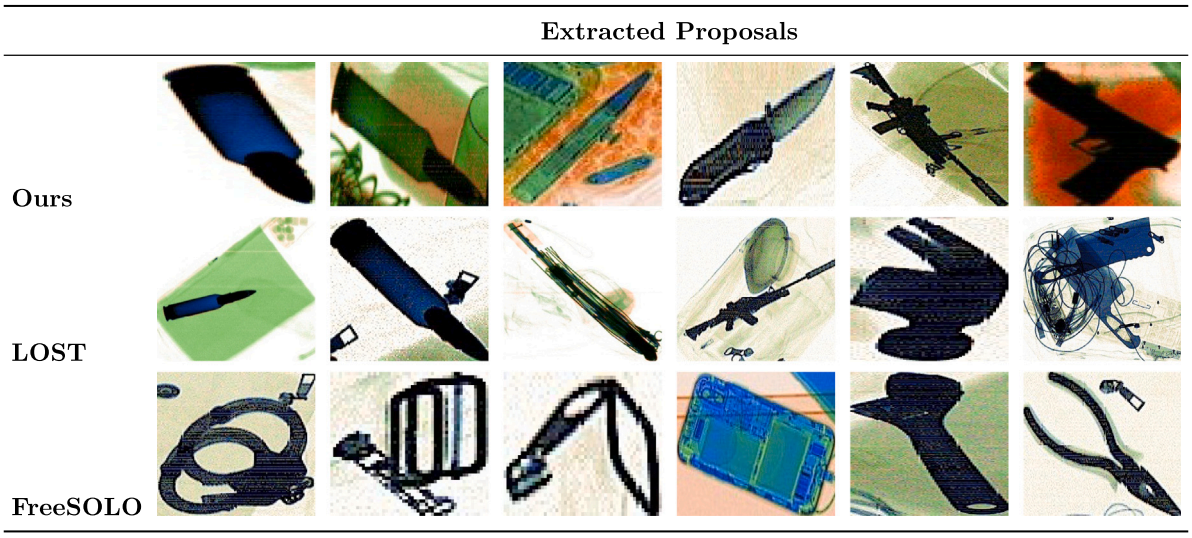

Fig. 5. Visual comparison of feature extraction modules across different clustering techniques. The **Ours**, **LOST**, and **FreeSOLO** methods demonstrate varying capabilities in detecting objects under cluttered conditions.

the framework improves the detection of threat objects in cluttered environments, increasing screening efficiency in high-security areas. Furthermore, the transferable feature extraction method extends the applicability of our approach beyond airport security, enabling its use in parcel inspections, customs enforcement, and critical infrastructure protection.

7.2. Theoretical implications

Our project presents several significant theoretical contributions to the field of X-ray image analysis. First, we introduce a novel teacher–student learning framework that refines multi-scale features by leveraging both global and local image crops, where, unlike prior approaches that relied on random local crops, we employ region proposals generated from earlier stages to guide the learning process. Second, we advance SSL in X-ray imagery by developing a structured, multi-stage pipeline that integrates contrastive multi-modal clustering with self-supervised distillation, enabling more robust feature representation. Lastly, we propose a unique fusion of zero-shot object detection with self-supervised representation learning, demonstrating improved threat object localization performance particularly in scenarios with scarce labeled data.

7.3. Advancements over existing research

Traditional supervised deep learning approaches for X-ray security rely heavily on large-scale annotated datasets. However, obtaining labeled security scans is highly resource-intensive due to:

1. The rarity of threat objects (e.g., firearms, explosives) in real-world data.
2. The need for expert human annotators, which is time-consuming and costly.
3. Privacy and security concerns, as sensitive airport screening data cannot always be freely shared for model training.

Compared to these supervised methods, X-SSL eliminates the need for manual annotations, making it highly scalable and adaptable. On the other hand, unsupervised threat detection methods have traditionally struggled with the complex grayscale nature of X-ray images and severe object occlusions. Prior approaches such as autoencoders or clustering-based feature extraction often fail to distinguish between threat and non-threat items due to the lack of semantic context in grayscale images. X-SSL overcomes these limitations by:

1. Incorporating multi-modal learning, where textual features improve the model's understanding of object semantics.
2. Using knowledge distillation, allowing a student model to refine features across multiple scales, improving threat localization robustness.

7.4. Limitations of LLMs

Despite the increasing utility of Large Language Models (LLMs) and multi-modal foundation models across vision–language tasks, their direct application to self-supervised X-ray image analysis, particularly in domains such as baggage threat localization, faces

several fundamental limitations. First, LLMs typically lack the visual grounding required to interpret the structural and semantic complexity of X-ray imagery, which is often grayscale, low in texture, and characterized by overlapping or occluded objects. Unlike natural images, X-rays demand domain-specific understanding and high spatial resolution, which most LLMs are not trained to accommodate (Wang, Li, & Chen, 2023; Zhang, Yang, & Li, 2023).

Furthermore, these models are predominantly pre-trained on large-scale natural image–text pairs (e.g., LAION (Schuhmann et al., 2022), COCO (Lin et al., 2014)) and thus suffer from poor domain transferability to specialized tasks without extensive fine-tuning or re-training (Ma, Liu, & Wong, 2023; Zhang et al., 2023). Their performance deteriorates further in zero-shot or low-shot settings, which are common in threat detection scenarios due to limited annotated data. Additionally, LLMs are inherently unsuitable for pixel-level tasks such as segmentation and anomaly detection, which require localized attention and fine-grained feature representations typically achieved via convolution or Transformer-based architectures optimized for spatial processing (Liu, Tian, & Wang, 2022).

From a safety and deployment perspective, LLMs pose significant challenges due to their opaque decision-making processes and limited interpretability, factors that are particularly problematic in high-stakes applications like security screening or medical diagnostics (Holzinger, Malle, Kieseberg, & Holzinger, 2022). Their substantial computational demands also hinder real-time deployment in operational environments. Lastly, current LLM-based frameworks are misaligned with unsupervised learning paradigms such as contrastive learning or clustering-based self-supervised methods, which have shown greater effectiveness in scenarios with limited or no labels (Tuli, Ramesh, & Ghassemi, 2023).

7.5. Limitations

While X-SSL presents a significant advancement, the study has several limitations that should be addressed in future research.

- **False Positives and Class Imbalance:** While the model improves threat localization performance, it still exhibits a high false positive rate, especially during multi-modal clustering stage. This includes, wrongfully classifying chargers, laptops or other similar electronic gadgets as threat items causing in decline of accuracy. Moreover, the datasets used in this study contain a limited number of rare threat types, leading to potential class imbalances. For future work, incorporating synthetic threat augmentation and hard negative mining could help reduce false positives and improve robustness.
- **Ethical and Legal Considerations:** The datasets used are publicly available but real-world security screening involves privacy-sensitive data. Deploying AI-based threat detection raises ethical concerns about false accusations and biases in AI-driven security decisions. Moreover, the study does not explicitly account for adversarial attacks, where an attacker intentionally modifies an object to evade detection. For future work, including adversarial defense strategies and bias mitigation techniques, is necessary before large-scale deployment.
- **Computational Constraints:** Although X-SSL eliminates the need for labeled data, its self-supervised knowledge distillation process is computationally intensive. It requires large-scale feature extraction, making it resource-heavy for on-device inference in low-power security systems. For future work, knowledge distillation-based model compression could make the training framework more practical for training process.

8. Conclusion

This paper introduces X-SSL, a novel self-supervised framework for automated threat detection in X-ray baggage images, addressing key challenges such as object occlusion, clutter, and the scarcity of annotated data. X-SSL integrates three core components: MaskCut for zero-shot spatial region extraction, contrastive multi-modal clustering to differentiate threat and non-threat objects, and a self-supervised knowledge distillation strategy that enhances feature learning through multi-scale teacher–student models. X-SSL leverages both visual and textual encoders to improve region proposal generation and classification, offering robust performance across diverse datasets. Our evaluation on benchmark datasets such as PIDray and CLCXray demonstrates that X-SSL significantly outperforms previous SOTA models in detection, segmentation, and classification tasks. Notably, X-SSL achieves superior accuracy and AP across challenging test subsets, reinforcing its ability to generalize effectively across varying X-ray scanner types and object categories. As one of the first frameworks to integrate multi-modal learning and self-supervision in X-ray threat detection, X-SSL highlights the potential of these approaches for enhancing security screening technologies. Future work will focus on expanding the framework for highly imbalanced anomaly detection and cross-domain generalization, with further exploration of more advanced multi-modal fusion strategies to handle rare and evolving threat scenarios.

Abbreviations

SOTA	state-of-the-art
GAE	Generalized Auto-Encoder
SSL	Self-Supervised Learning
CNN	Convolutional Neural Network
R-CNN	Region-based Convolutional Neural Network
ViT	Vision Transformer
ROI	Region of Interest
AP	Average Precision
mAP	mean Average Precision

CRediT authorship contribution statement

Yonathan Michael: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation, Conceptualization. **Mohamad Alansari:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation, Conceptualization. **Abdelfatah Ahmed:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation. **Naoufel Werghi:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Andreas Henschel:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis.

Funding

This work is supported by research funds from Khalifa University, No: CIRA2021-052, and the Advanced Technology Research Center Program (ASPIRE), No. AARE20-279

Declaration of competing interest

The authors declare that they have no competing interests.

Data availability

The datasets used in this work are publicly available and can be accessed through the following links: PIDray: https://github.com/lutao2021/PIDray CLCxray: https://github.com/GreysonPhoenix/CLCXray.

References

Ahmed, A., Alansari, M., Alnuaimi, K., Velayudhan, D., Hassan, T., & Werghi, N. (2023). Detection transformer framework for recognition of heavily occluded suspicious objects. In *2023 IEEE international conference on computational intelligence and virtual environments for measurement systems and applications* (pp. 1–6). http://dx.doi.org/10.1109/CIVEMSA57781.2023.10231015.

Ahmed, A., Obeid, A., Velayudhan, D., Hassan, T., Damiani, E., & Werghi, N. (2022). Balanced affinity loss for highly imbalanced baggage threat contour-driven instance segmentation. In *2022 IEEE international conference on image processing* (pp. 981–985). IEEE.

Ahmed, A., Velayudhan, D., ElMezain, M., Al Radi, M., Boudiaf, A., Hassan, T., Deriche, M., Bennamoun, M., & Werghi, N. (2024). CLIFS: CLIP-driven few-shot learning for baggage threat classification. In *2024 IEEE international conference on image processing* (pp. 753–759). IEEE.

Akcay, S., Atapour-Abarghouei, A., & Breckon, T. P. (2018). Ganomaly: Semi-supervised anomaly detection via adversarial training. arXiv:1805.06725.

Akçay, S., Atapour-Abarghouei, A., & Breckon, T. P. (2019). Skip-ganomaly: Skip connected and adversarially trained encoder-decoder anomaly detection. arXiv:1901.08954.

Akçay, S., Kundegorski, M. E., Willcocks, C. G., & Breckon, T. P. (2018). Using deep convolutional neural network architectures for object classification and detection within x-ray baggage security imagery. *IEEE Transactions on Information Forensics and Security*, *13*(9), 2203–2215.

Alansari, M., Ahmed, A., Alnuaimi, K., Velayudhan, D., Hassan, T., Javed, S., Bennamoun, M., & Werghi, N. (2024). Multi-scale hierarchical contour framework for detecting cluttered threats in baggage security. *IEEE Access*.

Andrews, J. T., Jaccard, N., Rogers, T. W., & Griffin, L. D. (2017). Representation-learning for anomaly detection in complex x-ray cargo imagery. In *Anomaly detection and imaging with x-rays (ADIX) II*: *vol. 10187*, (pp. 46–56). SPIE.

Cao, S., Joshi, D., Gui, L., & Wang, Y.-X. (2023). HASSOD: Hierarchical adaptive self-supervised object detection. In *NeurIPS*.

Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. In *Proceedings of the international conference on computer vision*.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16 × 16 words: Transformers for image recognition at scale. In *International conference on learning representations*. URL https://openreview.net/forum?id=YicbFdNTTy.

Fang, C., Liu, J., Han, P., et al. (2023). Few-shot threat detection for X-ray security images. arXiv:4069.

Griffin, L. D., Caldwell, M., Andrews, J. T., & Bohler, H. (2018). "Unexpected item in the bagging area": Anomaly detection in x-ray security images. *IEEE Transactions on Information Forensics and Security*, *14*(6), 1539–1553.

Hassan, T., Akçay, S., Bennamoun, M., & Khan, S. (2023). Unsupervised anomaly instance segmentation for baggage threat recognition. arXiv:2107.07333.

Hassan, T., Akcay, S., Bennamoun, M., Khan, S., & Werghi, N. (2022). Tensor pooling-driven instance segmentation framework for baggage threat recognition. *Neural Computing and Applications*, *34*(2), 1239–1250. http://dx.doi.org/10.1007/s00521-021-06411-x.

Hassan, T., Bettayeb, M., Akçay, S., Khan, S., Bennamoun, M., & Werghi, N. (2020). Detecting prohibited items in X-ray images: a contour proposal learning approach. In *2020 IEEE international conference on image processing* (pp. 2016–2020). IEEE.

Hassan, T., Shafay, M., Akcay, S., Khan, S., Bennamoun, M., Damiani, E., & Werghi, N. (2020). Meta-transfer learning driven tensor-shot detector for the autonomous localization and recognition of concealed baggage threats. *Sensors*.

He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask r-cnn. In *Proceedings of the IEEE international conference on computer vision* (pp. 2961–2969).

Holzinger, A., Malle, B., Kieseberg, P., & Holzinger, K. (2022). Information fusion and explainability for trustworthy AI in security and health. *Information Fusion*, *82*, 432–450. http://dx.doi.org/10.1016/j.inffus.2021.12.009.

Hong, S., Zhou, Y., & Xu, W. (2025). FEA-DETR: An enhanced ConvNet for detecting prohibited objects in X-Ray images using frequency and edge aware information. In *ICASSP 2025-2025 IEEE international conference on acoustics, speech and signal processing* (pp. 1–5). IEEE.

Hu, Y., Chen, Y., Huang, R., Qin, Y., & Zheng, Q. (2024). A hierarchical convolutional model for biomedical relation extraction. *Information Processing & Management*, *61*(1), Article 103560.

Kerdvibulvech, C. (2025). Multimodal AI model for zero-shot vehicle brand identification. *Multimedia Tools and Applications*, 1–20.

Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft coco: Common objects in context. In *Computer vision–ECCV 2014: 13th European conference, Zurich, Switzerland, September 6-12, 2014, proceedings, part v 13* (pp. 740–755). Springer.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., & Guo, B. (2021). Swin transformer: Hierarchical vision transformer using shifted windows. URL https://arxiv.org/abs/2103.14030.

Liu, Q., Tian, Y., & Wang, Y. (2022). Self-supervised learning for medical imaging: A survey and perspective. *Medical Image Analysis*, *77*, Article 102355. http://dx.doi.org/10.1016/j.media.2022.102355.

Ma, Y., Liu, J., & Wong, K. (2023). Failure cases of multimodal foundation models in X-ray vision tasks. arXiv preprint arXiv:2310.01456.

Miao, C., Xie, L., Wan, F., Su, C., Liu, H., Jiao, J., & Ye, Q. (2019). Sixray: A large-scale security inspection x-ray benchmark for prohibited item discovery in overlapping images. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 2119–2128).

Nasim, A., Akram, M. U., Khan, A. M., et al. (2024). A cross-domain threat screening and localization framework using self-supervised learning. arXiv:10677838.

National Research Council (1996). Airline passenger security screening: New technologies and implementation issues. The National Academics Press.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. arXiv:2103.00020.

Ren, S., He, K., Girshick, R., & Sun, J. (2015). Faster r-cnn: Towards real-time object detection with region proposal networks. *Advances in Neural Information Processing Systems*, *28*.

Saavedra, D., Banerjee, S., & Mery, D. (2021). Detection of threat objects in baggage inspection with X-ray images using deep learning. *Neural Computing and Applications*, *33*, 7803–7819.

Schuhmann, C., Beaumont, R., Vencu, R., Gordon, C., Wightman, R., Cherti, M., Coombes, T., Katta, A., Mullis, C., Wortsman, M., et al. (2022). Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in Neural Information Processing Systems*, *35*, 25278–25294.

Shafay, M., Hassan, T., Ahmed, A., Velayudhan, D., Dias, J., & Werghi, N. (2022). Programmable broad learning system to detect concealed and imbalanced baggage threats. In *2022 2nd international conference on digital futures and transformative technologies* (pp. 1–6). IEEE.

Sigman, J. B., Spell, G. P., Liang, K. J., et al. (2020). Background adaptive faster R-CNN for semi-supervised convolutional object detection of threats in X-ray images. arXiv:2010.01202.

Siméoni, O., Puy, G., Vo, H. V., Roburin, S., Gidaris, S., Bursuc, A., Pérez, P., Marlet, R., & Ponce, J. (2021). Localizing objects with self-supervised transformers and no labels. In *Proceedings of the British machine vision conference*.

Singh, A., & Dhiraj (2024). Advancements in machine learning techniques for threat item detection in X-ray images: A comprehensive survey. arXiv:00348-2.

Tao, R., Wei, Y., Li, H., Liu, A., Ding, Y., Qin, H., & Liu, X. (2021). Over-sampling de-occlusion attention network for prohibited items detection in noisy X-ray images. arXiv preprint arXiv:2103.00809.

Tuli, A., Ramesh, S., & Ghassemi, M. (2023). Are multimodal foundation models effective for medical imaging? arXiv preprint arXiv:2307.09008.

Uroukov, I., & Speller, R. (2015). A preliminary approach to intelligent X-ray imaging for baggage inspection at airports. *Signal Processing Research*, *4*, 1–11.

Velayudhan, D., Ahmed, A., Hassan, T., Owais, M., Gour, N., Bennamoun, M., Damiani, E., & Werghi, N. (2024). Semi-supervised contour-driven broad learning system for autonomous segmentation of concealed prohibited baggage items. *Visual Computing for Industry, Biomedicine, and Art*, *7*(1), 30.

Velayudhan, D., Hassan, T., Ahmed, A. H., Damiani, E., & Werghi, N. (2022). Baggage threat recognition using deep low-rank broad learning detector. In *2022 IEEE 21st mediterranean electrotechnical conference* (pp. 966–971).

Velayudhan, D., Hassan, T., Damiani, E., & Werghi, N. (2022). Recent advances in baggage threat detection: A comprehensive and systematic survey. *ACM Computing Surveys*.

Velayudhan, D., Hassan Ahmed, A., Hassan, T., Bennamoun, M., Damiani, E., & Werghi, N. (2022). Transformers for imbalanced baggage threat recognition. In *2022 IEEE international symposium on robotic and sensors environments* (pp. 1–7). http://dx.doi.org/10.1109/ROSE56499.2022.9977427.

Wang, X., Girdhar, R., Yu, S. X., & Misra, I. (2023). Cut and learn for unsupervised object detection and instance segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3124–3134).

Wang, J., Li, X., & Chen, H. (2023). Challenges in applying large vision models to medical imaging: A review. *Medical Image Analysis*, *87*, Article 102833. http://dx.doi.org/10.1016/j.media.2023.102833.

Wang, Y., Shen, X., Hu, S. X., Yuan, Y., Crowley, J. L., & Vaufreydaz, D. (2022). Self-supervised transformers for unsupervised object discovery using normalized cut. In *Conference on computer vision and pattern recognition*.

Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., & Shao, L. (2021). Pyramid vision transformer: A versatile backbone for dense prediction without convolutions. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 568–578).

Wang, W., Xie, E., Li, X., Fan, D.-P., Song, K., Liang, D., Lu, T., Luo, P., & Shao, L. (2022). PVT v2: Improved baselines with pyramid vision transformer. *Computational Visual Media*, [ISSN: 2096-0662] *8*(3), 415–424. http://dx.doi.org/10.1007/s41095-022-0274-8.

Wang, X., Yu, Z., Mello, S. D., Kautz, J., Anandkumar, A., Shen, C., & Alvarez, J. M. (2022). FreeSOLO: Learning to segment objects without annotations. arXiv:2202.12181.

Wang, B., Zhang, L., Wen, L., Liu, X., & Wu, Y. (2021). Towards real-world prohibited item detection: A large-scale X-ray benchmark. In *Proceedings of the IEEE/CVF international conference on computer vision* (pp. 5412–5421).

Wei, Y., Tao, R., Wu, Z., Ma, Y., Zhang, L., & Liu, X. (2020). Occluded prohibited items detection: An X-ray security inspection benchmark and de-occlusion attention module. In *Proceedings of the 28th ACM international conference on multimedia* (pp. 138–146).

Wu, Y., Qiu, L., Wang, J., & Feng, S. (2025). The use of convolutional neural networks for abnormal behavior recognition in crowd scenes. *Information Processing & Management*, *62*(1), Article 103880.

Xia, R., Li, G., Huang, Z., et al. (2021). Classify and localize threat items in X-ray imagery with multiple attention mechanisms and high-resolution and high-semantic features. arXiv:9513650.

Xiao, Y., Zhou, G., Xie, Z., Liu, J., & Huang, J. X. (2024). Learning dual disentangled representation with self-supervision for temporal knowledge graph reasoning. *Information Processing & Management*, *61*(3), Article 103618.

Zhan, M., Shi, X., Liu, F., & Hu, R. (2023). IGCNN-FC: Boosting interpretability and generalization of convolutional neural networks for few chest X-rays analysis. *Information Processing & Management*, *60*(3), Article 103258.

Zhang, Z., Yang, Y., & Li, K. (2023). Limitations of CLIP in medical imaging: A case study on chest X-rays. arXiv preprint arXiv:2303.09667.

Zhao, C., Zhu, L., Dou, S., Deng, W., & Wang, L. (2022). Detecting overlapped objects in X-ray security imagery by a label-aware mechanism. *IEEE Transactions on Information Forensics and Security*, *17*, 998–1009. http://dx.doi.org/10.1109/TIFS.2022.3154287.